

EXPLICIT EXTREME VALUES OF THE ARGUMENT OF THE RIEMANN ZETA-FUNCTION

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ABSTRACT. We investigate explicit extreme values of the argument of the Riemann zeta-function in short intervals. As an application, we improve the result of Conrey and Turnage-Butterbaugh concerning r -gaps between zeros of the Riemann zeta-function.

1. Introduction and statement of results

The argument of the Riemann zeta-function ζ on the critical line, usually denoted as $S(t)$, is a fascinating and intricate aspect of one of the most celebrated functions in number theory. The function is defined by $S(t) := \frac{1}{\pi} \arg \zeta(\frac{1}{2} + it) = \frac{1}{\pi} \operatorname{Im} \int_{\infty}^{1/2} \frac{\zeta'}{\zeta}(\alpha + it) d\alpha$ if t is equal to neither 0 nor the imaginary part of a zero of ζ . If t is equal to 0 or the imaginary part of a zero, then $S(t) = (S(t+0) + S(t-0))/2$. By the argument principle, this function is influenced by the distribution of zeros of the Riemann zeta-function. The relationship is visualized by the Riemann-von Mangoldt formula:

$$N(T) = \frac{T}{2\pi} \log \left(\frac{T}{2\pi e} \right) + \frac{7}{8} + S(T) + O\left(\frac{1}{T}\right),$$

where $N(T)$ denotes the number of zeros $\rho = \beta + i\gamma$ satisfying $0 < \gamma < T$ of ζ counted with multiplicity. If T is equal to the imaginary part of a zero, $N(T) = (N(T+0) + N(T-0))/2$.

In this paper, we discuss extreme values of $S(t+h) - S(t)$. The values of $S(t+h) - S(t)$ capture information on the number of zeros, as expressed by

$$N(T+h) - N(T) = \frac{h}{2\pi} \log T + S(T+h) - S(T) + O\left(h + \frac{1}{T}\right). \quad (1.1)$$

Since detailed information about the zeros of the Riemann zeta-function has rich applications to the prime numbers, the study of $S(t+h) - S(t)$ is therefore also important. For this object, Selberg showed in an unpublished work that there exists a positive number $c = c(a, b)$, depending on arbitrary absolute positive constants a, b , such that for any large T and any $h \in [a(\log T)^{-1}, b(\log \log T)^{-1}]$,

$$\sup_{t \in [T, 2T]} \{\pm(S(t+h) - S(t))\} \geq c(h \log T)^\eta \quad (1.2)$$

holds with $\eta = 1/2$ under the Riemann Hypothesis (RH). Later, Tsang gave a proof of this result in [13] and also proved that (1.2) holds with $\eta = 1/3$ unconditionally. More recently, the first author in [8] used his results to study the distribution of zeros. In the paper, under RH, the first author also considered the explicit extreme values of $S(t+h) - S(t)$ and showed that $c = c(a, b)$ in inequality (1.2) can be calculated by (4.1) in [8, §4], which is essentially obtained by following the argument of Selberg/Tsang straightforwardly. With this approach, one inevitably has $c < 1/\sqrt{2e\pi}$ when a is large, and b is small. The aim of this paper is to

2020 *Mathematics Subject Classification.* 11M06; 11M26.

Key words and phrases. the Riemann zeta-function, gaps of zeros, extreme values, the resonance method.

improve the explicit extreme values by using the method of Montgomery-Odlyzko [9]. The first result is the following.

Theorem 1. *Assume RH. For any large T and any $h \in [C/\log T, c/\log \log T]$ with positive constants C large and c small, we have*

$$\sup_{T \leq t \leq 2T} \{\pm(S(t+h) - S(t))\} \geq (1-E) \sqrt{\frac{h}{\pi} \log T},$$

where the error term E satisfies

$$E \ll \sqrt{h \log \log T} + \min \left\{ \sqrt{\frac{\log^3(h \log T)}{h \log T}}, \frac{(\log \log T)^{3/2}}{h^{3/2} \log T} \right\}.$$

This theorem can be applied to evaluate gaps of zeros of the Riemann zeta-function. Let $0 < \gamma_1 \leq \gamma_2 \leq \dots \leq \gamma_n \leq \dots$ denote the sequence of ordinates of the zeros of ζ in the upper half plane. We define the normalized large/small r -gap of nontrivial zeros by

$$\lambda_r = \limsup_{n \rightarrow +\infty} \frac{\gamma_{n+r} - \gamma_n}{2\pi r / \log \gamma_n}, \quad \mu_r = \liminf_{n \rightarrow +\infty} \frac{\gamma_{n+r} - \gamma_n}{2\pi r / \log \gamma_n}.$$

From the Riemann-von Mangoldt formula, we have the trivial bounds $\mu_r \leq 1 \leq \lambda_r$. The non-trivial bounds in the case $r = 1$ have been studied by many mathematicians. The current best bounds under RH are $\lambda_1 > 3.18$ by Bui-Milinovich [3] and $\mu_1 < 0.515396$ by Preobrazhenskii [10]. For general r , Selberg [11, p.355] announced the nontrivial bounds of λ_r, μ_r of the form

$$\lambda_r \geq 1 + \frac{\Theta}{r^\alpha}, \quad \mu_r \leq 1 - \frac{\vartheta}{r^\alpha} \tag{1.3}$$

for all positive integer r . The numbers Θ, ϑ which may depend on r are greater than some absolute positive constants. Here, we may take α as $2/3$ unconditionally, and as $1/2$ under RH. Recently, Conrey and Turnage-Butterbaugh [5] proved an explicit result for the conditional bound. Specifically, they showed that (1.3) holds for $\Theta = 0.599648$ and $\vartheta = 0.379674$ with $\alpha = 1/2$ uniformly for $r \geq 1$ under RH. These results have been improved to $\Theta = A_0 := \max_{B>0} \frac{2B}{\pi} \arctan\left(\frac{\pi}{B^2}\right) = 0.9064997\dots$, and $\vartheta = 0.484604$ in [8]. Moreover, Conrey and Turnage-Butterbaugh proved that $\Theta = \vartheta = A_0 + o(1)$ as $r \rightarrow +\infty$. As a consequence of Theorem 1 combined with the Riemann-von Mangoldt formula, we can improve the constant A_0 to $\sqrt{2} = 1.4142\dots$.

Theorem 2. *Assume RH. For any sufficiently large r , we have*

$$\lambda_r \geq 1 + \frac{\sqrt{2}}{\sqrt{r}} - C_1 \frac{(\log r)^{3/2}}{r}, \quad \mu_r \leq 1 - \frac{\sqrt{2}}{\sqrt{r}} + C_2 \frac{(\log r)^{3/2}}{r}. \tag{1.4}$$

Here, C_1 and C_2 are some absolute positive constants.

To prove Theorem 1, we employ the resonance method established by Soundararajan [12]. From the celebrated works due to Soundararajan [12], and Bondarenko and Seip [1], the resonance method is now widely regarded as a powerful tool for detecting extreme values of number-theoretic objects. Inspired by these studies, we apply the resonance method to $S(t+h) - S(t)$. The method due to Montgomery and Odlyzko is a precursor of our approach. In fact, by combining our method using the resonance method with the Riemann-von Mangoldt formula, we can recover the original result of Montgomery and Odlyzko.

In [9], Montgomery and Odlyzko showed that $\lambda_1 > 1.9799$ and $\mu_1 < 0.5179$. For a large real number T and $L \leq T/(\log T)^2$, they studied the function τ defined by

$$\tau(\xi; f) := \xi - \left(\operatorname{Re} \frac{2}{\pi} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin(\pi \xi \frac{\log k}{\log T}) f(m) \overline{f(km)} \right) / \sum_{n \leq L} |f(n)|^2,$$

where ξ is a positive number, and f is a certain arithmetic function. They showed (for the case $r = 1$ and extended $r \geq 2$ by Conrey and Turnage-Butterbaugh [5]) that if there exists ξ_r such that $\tau(\xi_r; f) < r$, then $\lambda_r \geq \xi_r$, and if there exists ξ_r such that $\tau(\xi_r; f) > r$, then $\mu_r \leq \xi_r$.

To conclude this section, we give a limitation of the method of Montgomery and Odlyzko.

Theorem 3. *Let f be an arithmetic function not identically zero, let L be large, and let $h > 0$. For any $W > 0$, we have*

$$\begin{aligned} & \left| \operatorname{Re} \frac{2}{\pi} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin(\frac{h}{2} \log k) f(m) \overline{f(km)} \right| / \sum_{n \leq L} |f(n)|^2 \\ & \leq \max_{1 \leq l \leq L} \left\{ \frac{\sqrt{W}}{2} \varphi\left(\frac{h}{2\pi} \log(L/l)\right) + \frac{h \log l}{\pi \sqrt{W}} \right\} + O(h\sqrt{W}). \end{aligned} \quad (1.5)$$

Here, $\varphi(x) = \int_0^x (\sin(\pi u)/\pi u)^2 du$. In particular, we have

$$|\tau(\xi; f) - \xi| \leq \max_{1 \leq l \leq L} \left\{ \frac{\sqrt{W}}{2} \varphi\left(\xi \frac{\log(L/l)}{\log T}\right) + \frac{2\xi}{\sqrt{W}} \frac{\log l}{\log T} \right\} + O\left(\frac{\xi}{\log T} \sqrt{W}\right)$$

for any $\xi > 0$, any large L, T , and any $W > 0$.

By this theorem together with a numerical calculation, we obtain that the limitations of large/small gaps of zeros in the method of Montgomery-Odlyzko are $\lambda_1 \geq 3.022$, $\mu_1 \leq 0.508$ when $L \leq T$. There are previous works on such limitations in the method [4], [6]. Conrey, Ghosh, and Gonek [4] showed that $\lambda_1 \geq 3.74$ and $\mu_1 < 1/2$ are limitations of the Montgomery-Odlyzko method. Moreover, the recent work by Goldston, Trudgian, and Turnage-Butterbaugh [6] improved on the result on their limitation of small gaps to $\mu_1 \leq 0.5042$. They also remarked that their method can be applied to the bound of λ_1 , which leads us to $\lambda_1 \geq 3.6747$. Our result gives an improvement on those results. On the other hand, Bui and Milinovich [3] applied Hall's method [7] to prove $\lambda_1 > 3.18$ under RH. Our limitation for λ_1 shows that the result of Bui-Milinovich goes beyond the barrier imposed by the method of Montgomery-Odlyzko. Furthermore, we can also see that $\lambda_r \geq 1 + \sqrt{2/r} - O(1/r)$, $\mu_r \leq 1 - \sqrt{2/r} + O(1/r)$ are limitations of their method for r -gaps of zeros when $L \leq T$. This observation shows that the constant $\sqrt{2}$ in Theorem 2 is optimal.

This paper is organized as follows. In Section 2 we discuss the relationship between large values of $S(t)$ in short intervals and the gaps between consecutive r zeros of the Riemann zeta-function. In Section 3, we apply the resonance method to $S(t)$ in short intervals. Combining this result and Proposition 4, we prove Theorem 1 in Section 4. In Section 5, we prove Theorem 2 by using Theorem 1 and the relationship between $S(t)$ and gaps of zeros established in Section 2. In Section 6, we prove Theorem 3, and finally in Section 7, we derive the resulting limitations on large and small gaps between zeros that follow from Theorem 3.

2. A relationship between $S(t)$ and large/small gaps of consecutive r zeros

By the same strategy as in the proof of Theorem 1 in [8], we obtain the following relation between $S(t)$ and gaps of zeros.

Proposition 1. *Let r be a positive integer, and let θ be a positive number which may depend on r . Then the inequality $\lambda_r > \theta$ holds if and only if there exist numbers $b > 0$, $\theta' > 1$ and a sequence $\{T_n\}$ satisfying $\theta' > \theta$, $b > r(\theta' - 1)$, and $T_n \rightarrow +\infty$ as $n \rightarrow +\infty$ such that*

$$\inf_{t \in [T_n, 2T_n]} \{S(t + 2\pi r\theta' / \log T_n) - S(t)\} \leq -b.$$

Similarly, the inequality $\mu_r < \theta$ holds if and only if there exist numbers $b > 0$, $0 < \theta' < 1$ and a sequence $\{T_n\}$ satisfying $\theta' < \theta$, $b > r(1 - \theta')$, and $T_n \rightarrow +\infty$ as $n \rightarrow +\infty$ such that

$$\sup_{t \in [T_n, 2T_n]} \{S(t + 2\pi r\theta' / \log T_n) - S(t)\} \geq b.$$

Proof. Since the first and second assertion can be proved by the same argument, we only give the proof of the first assertion. We use the simple equivalence which is that, for any $\{T_n\}$ satisfying $T_n \rightarrow +\infty$ as $n \rightarrow +\infty$, there exists some $t \in [T_n, 2T_n]$ such that

$$N(t + h) - N(t) < r \tag{2.1}$$

if and only if the inequality

$$\sup_{\gamma_m, \gamma_{m+r} \in [T_n, 2T_n+h]} \frac{\gamma_{m+r} - \gamma_m}{h} > 1 \tag{2.2}$$

holds.

First, we assume $\lambda_r > \theta$. Then there exist a number θ' and a sequence $\{T_n\}$ satisfying $\theta' > \theta$ and $T_n \rightarrow +\infty$ as $n \rightarrow +\infty$ such that

$$\sup_{\gamma_m, \gamma_{m+r} \in [T_n, 2T_n]} \frac{\gamma_{m+r} - \gamma_m}{2\pi r\theta' / \log T_n} \geq \sup_{\gamma_m, \gamma_{m+r} \in [T_n, 2T_n]} \frac{\gamma_{m+r} - \gamma_m}{2\pi r\theta' / \log(\gamma_m/2)} > 1$$

holds for any sufficiently large n . Therefore, (2.2) holds when $h = 2\pi r\theta' / \log T_n$ and n is sufficiently large. Hence, there exists a $t \in [T_n, 2T_n]$ such that (2.1) holds with $h = 2\pi r\theta' / \log T_n$, which is also equivalent to

$$N(t + 2\pi r\theta' / \log T_n) - N(t) \leq r - 1/2.$$

Combining this with (1.1), we have

$$\begin{aligned} \inf_{t \in [T_n, 2T_n]} (S(t + 2\pi r\theta' / \log T_n) - S(t)) &\leq \inf_{t \in [T_n, 2T_n]} (S(t + 2\pi r\theta' / \log T_n) - S(t)) \\ &\leq r - 1/2 - r\theta' + o(1) \leq -b \end{aligned}$$

with $b = r(\theta' - 1) + 1/3$.

Next, we assume that there exist numbers b, θ' and a sequence $\{T_n\}$ satisfying $\theta' > \theta$, $b > r(\theta' - 1)$, and $T_n \rightarrow +\infty$ as $n \rightarrow +\infty$ such that

$$\inf_{t \in [T_n, 2T_n]} (S(t + 2\pi r\theta' / \log T_n) - S(t)) \leq -b$$

for any sufficiently large n . Then, it holds by (1.1) that for any large n

$$N(t + 2\pi r\theta' / \log T_n) - N(t) \leq r\theta' \frac{\log t}{\log T_n} - b + o(1) \leq r - (b - r(\theta' - 1)) + o(1)$$

for some $t \in [T_n, 2T_n]$. Therefore, (2.1) holds when $h = 2\pi r\theta'/\log T_n$ and n is sufficiently large. Hence, we find that

$$\begin{aligned}\lambda_r &= \limsup_{m \rightarrow +\infty} \frac{\gamma_{m+r} - \gamma_m}{2\pi r / \log \gamma_m} \geq \lim_{n \rightarrow +\infty} \sup_{\gamma_m, \gamma_{m+r} \in [T_n, 2T_n+h]} \frac{\gamma_{m+r} - \gamma_m}{2\pi r / \log \gamma_m} \\ &= \lim_{n \rightarrow +\infty} \sup_{\gamma_m, \gamma_{m+r} \in [T_n, 2T_n+h]} \frac{\gamma_{m+r} - \gamma_m}{h} \frac{h}{2\pi r / \log T_n} \frac{\log \gamma_m}{\log T_n} \geq \theta' > \theta,\end{aligned}$$

which completes the proof of Proposition 1. $\square$

Remark 1. In Proposition 1, if we change the interval $[T_n, 2T_n]$ to $[T_n^a, 2T_n]$ for some $0 < a < 1$, then the equivalence no longer holds. Although the statement can be suitably modified, the resulting inequalities for λ_r, μ_r become weaker than the original form. Hence, we consider the extreme value of $S(t+h) - S(t)$ over the interval $[T, 2T]$ in Theorem 1.

3. Resonance method

The resonance method aims to extract the large values of an objective function by comparing the mean value of the objective function multiplied by a “resonator” with that of the resonator itself. In this paper, we take the resonator to be the Dirichlet polynomial

$$R(t) = \sum_{n \leq L} f(n)n^{-it},$$

following the works [1], [2], [9], and [12]. Here, the arithmetic function f is chosen suitably depending on the objective function. In this section, we evaluate the extreme value of $S(t+h) - S(t)$ by means of general forms of resonators. We construct a suitable resonator for our purpose in Section 4.

In this section, we aim to prove the following proposition.

Proposition 2. *Assume RH. For any arithmetic function f that is not identically zero, any large L, T satisfying $L \leq T/(\log T)^2$, and any $h > 0$ we have*

$$\begin{aligned}&\sup_{T \leq t \leq 2T} \{\pm(S(t+h) - S(t))\} \\ &\geq \mp \left(1 + O\left(\frac{1}{T}\right)\right) \frac{2}{\pi} \operatorname{Re} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) f(m) \overline{f(km)} \Big/ \sum_{n \leq L} |f(n)|^2 + O\left(\frac{1}{T}\right).\end{aligned}$$

Here, the implicit constant is absolute.

3.1. Preliminaries. Throughout this paper, we set $\Phi(t) = e^{-t^2/2}$. As an auxiliary result, we first prove the following proposition.

Proposition 3. *Assume RH. For any arithmetic function f , any $L, T \geq 3$ satisfying $L \leq T/(\log T)^2$, and any $h > 0$ we have*

$$\begin{aligned}&\int_{-\infty}^{\infty} \{S(t + \frac{h}{2}) - S(t - \frac{h}{2})\} |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \\ &= -\sqrt{2\pi} \frac{T}{\log T} \frac{2}{\pi} \operatorname{Re} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) f(m) \overline{f(km)} + O\left(\frac{1}{T} \sum_{n \leq L} |f(n)|^2\right).\end{aligned}$$

To show this proposition, we require some auxiliary lemmas.

Lemma 3.1. *For any arithmetic function f and any $L \geq 3$ we have*

$$|R(t)|^2 \leq \sum_{m,n \leq L} |f(m)f(n)| \leq L \sum_{n \leq L} |f(n)|^2. \quad (3.1)$$

Proof. The first inequality of (3.1) is obvious by the triangle inequality. We also find by the Cauchy-Schwarz inequality that

$$\sum_{m,n \leq L} |f(m)f(n)| = \left(\sum_{n \leq L} |f(n)| \right)^2 \leq \sum_{n \leq L} 1 \times \sum_{n \leq L} |f(n)|^2 = L \times \sum_{n \leq L} |f(n)|^2.$$

Hence, we obtain inequality (3.1). $\square$

The following lemma gives an explicit bound for estimates shown in Lemma 5 of [2].

Lemma 3.2. *For any arithmetic function f and any $L, T \geq 3$ satisfying $L \leq T/(\log T)^2$ we have*

$$\int_{-\infty}^{\infty} |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt = \sqrt{2\pi} \frac{T}{\log T} \left(1 + O\left(\frac{1}{T}\right)\right) \sum_{n \leq L} |f(n)|^2.$$

Proof. It holds from the definition of $R(t)$ that

$$\int_{-\infty}^{\infty} |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt = \sqrt{2\pi} \frac{T}{\log T} \sum_{m,n \leq L} f(m) \overline{f(n)} \left(\frac{m}{n}\right)^{-3iT/2} \Phi\left(\frac{T}{\log T} \log \frac{m}{n}\right)$$

since $\int_{-\infty}^{\infty} \Phi(u) e^{-ixu} du = \sqrt{2\pi} \Phi(x)$. Using (3.1), we find that

$$\begin{aligned} \left| \sum_{\substack{m,n \leq L \\ m \neq n}} f(m) \overline{f(n)} \left(\frac{m}{n}\right)^{3iT/2} \Phi\left(\frac{T}{\log T} \log \frac{m}{n}\right) \right| &\leq \Phi\left(\frac{T}{\log T} \frac{(\log T)^2}{2T}\right) \sum_{m,n \leq L} |f(m)f(n)| \\ &\leq \Phi\left(\frac{1}{2} \log T\right) L \sum_{n \leq L} |f(n)|^2 \leq \frac{1}{T} \sum_{n \leq L} |f(n)|^2. \end{aligned}$$

Adding the diagonal-terms to this, we complete the proof of Lemma 3.2. $\square$

Lemma 3.3. *Let V be an analytic function in the horizontal strip $\{z \in \mathbb{C} : -\frac{3}{2} \leq \operatorname{Im} z \leq 0\}$ satisfying $\sup_{-\frac{3}{2} \leq y \leq 0} |V(x + iy)| \ll (|x| \log^2 x)^{-1}$. For any $v \in \mathbb{R}$, we have*

$$\begin{aligned} &\int_{-\infty}^{\infty} \log \zeta\left(\frac{1}{2} + i(t + v)\right) V(t) dt \\ &= \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^{\frac{1}{2} + iv} \log n} \widehat{V}\left(\frac{\log n}{2\pi}\right) + 2\pi \sum_{\beta > \frac{1}{2}} \int_0^{\beta - \frac{1}{2}} V(\gamma - v - i\sigma) d\sigma - 2\pi \int_0^{\frac{1}{2}} V(-v - i\sigma) d\sigma. \end{aligned}$$

Here, $\widehat{V}$ is the Fourier transform of V defined by $\widehat{V}(z) = \int_{-\infty}^{\infty} V(x) e^{-2\pi i x z} dx$.

Proof. This is obtained by (2.14) in [13] and the argument below (2.14). $\square$

3.2. Proof of Proposition 3. We write

$$\begin{aligned} & \int_{-\infty}^{\infty} \log \zeta\left(\frac{1}{2} + i(t \pm \frac{h}{2})\right) |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \\ &= \sum_{m, n \leq L} f(m) \overline{f(n)} \int_{-\infty}^{\infty} \log \zeta\left(\frac{1}{2} + i(t \pm \frac{h}{2})\right) \left(\frac{m}{n}\right)^{-it} \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt. \end{aligned}$$

We use Lemma 3.3 with $V_{m,n}(z) = (n/m)^{iz} \Phi((z - 3T/2)/(T/\log T))$ to find that this equals

$$\begin{aligned} \sum_{m, n \leq L} f(m) \overline{f(n)} & \left\{ \sum_{k=2}^{\infty} \frac{\Lambda(k)}{k^{\frac{1}{2} \pm ih/2} \log k} \widehat{V_{m,n}}\left(\frac{\log k}{2\pi}\right) \right. \\ & \left. - 2\pi \int_0^{\frac{1}{2}} \left(\frac{m}{n}\right)^{-\sigma \pm i(h/2)} \Phi\left(\frac{\mp(h/2) - i\sigma - 3T/2}{T/\log T}\right) d\sigma \right\} \end{aligned}$$

under RH. The latter term is

$$\ll \sum_{m, n \leq L} |f(m)f(n)| \times L^{1/2} \Phi(\log T) \ll \frac{1}{T} \sum_{n \leq L} |f(n)|^2$$

by (3.1). Also, the former term is

$$= \sqrt{2\pi} \frac{T}{\log T} \sum_{m, n \leq L} \sum_{k=2}^{\infty} f(m) \overline{f(n)} \frac{\Lambda(k)}{\sqrt{k} \log k} k^{\mp ih/2} \left(\frac{km}{n}\right)^{-3iT/2} \Phi\left(\frac{T}{\log T} \log\left(\frac{km}{n}\right)\right)$$

since $\int_{-\infty}^{\infty} \Phi(u) e^{-ixu} du = \sqrt{2\pi} \Phi(x)$. Simple calculations using (3.1) show that

$$\begin{aligned} & \sum_{m, n \leq L} \sum_{\substack{2 \leq k \leq T^2 \\ km \neq n}} |f(m)f(n)| \frac{1}{k^{1/2}} \Phi\left(\frac{T}{\log T} \log\left(\frac{km}{n}\right)\right) \\ & \ll L \sum_{n \leq L} |f(n)|^2 \sum_{2 \leq k \leq T^2} \frac{1}{k^{1/2}} \Phi(\log T) \ll \frac{1}{T^2} \sum_{n \leq L} |f(n)|^2, \end{aligned}$$

and that

$$\begin{aligned} & \sum_{m, n \leq L} \sum_{\substack{k > T^2 \\ km \neq n}} |f(m)f(n)| \frac{1}{k^{1/2}} \Phi\left(\frac{T}{\log T} \log\left(\frac{km}{n}\right)\right) \\ & \ll \sum_{m, n \leq L} |f(m)f(n)| \sum_{k > T^2} \frac{1}{k^{1/2}} k^{-T/2 \log T} \ll L \sum_{n \leq L} |f(n)|^2 \times T^{-3} \ll \frac{1}{T^2} \sum_{n \leq L} |f(n)|^2. \end{aligned}$$

Following these, we have

$$\begin{aligned} & \int_{-\infty}^{\infty} \log \zeta\left(\frac{1}{2} + i(t \pm \frac{h}{2})\right) |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \\ &= \sqrt{2\pi} \frac{T}{\log T} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} k^{\mp ih/2} f(m) \overline{f(km)} + O\left(\frac{1}{T} \sum_{n \leq L} |f(n)|^2\right). \end{aligned}$$

This also leads to

$$\begin{aligned} & \int_{-\infty}^{\infty} \{S(t + \tfrac{h}{2}) - S(t - \tfrac{h}{2})\} |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \\ &= -\sqrt{2\pi} \frac{T}{\log T} \frac{2}{\pi} \operatorname{Re} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin(\tfrac{h}{2} \log k) f(m) \overline{f(km)} + O\left(\frac{1}{T} \sum_{n \leq L} |f(n)|^2\right). \end{aligned}$$

Thus, we complete the proof of Proposition 3. $\square$

3.3. Proof of Proposition 2. Let L, T be large numbers that satisfy $L \leq T/(\log T)^2$, and let $h > 0$. Those parameters are chosen later. Write

$$I = \int_{-\infty}^{\infty} \{S(t + \tfrac{h}{2}) - S(t - \tfrac{h}{2})\} |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt.$$

We then find by Proposition 3 that

$$I = -\frac{\sqrt{2\pi}T}{\log T} \frac{2}{\pi} \operatorname{Re} \sum_{2 \leq k \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin(\tfrac{h}{2} \log k) \sum_{km \leq L} f(m) \overline{f(km)} + O\left(\frac{1}{T} \sum_{n \leq L} |f(n)|^2\right). \quad (3.2)$$

First, we show that

$$\begin{aligned} I &= \int_{4T/3 \leq t \leq 5T/3} (S(t + \tfrac{h}{2}) - S(t - \tfrac{h}{2})) |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \\ &\quad + O\left(\frac{1}{T} \sum_{n \leq L} |f(n)|^2\right). \end{aligned} \quad (3.3)$$

Using Lemma 3.1 and the estimate $S(t) \ll \log(|t| + 3)$, we find by simple calculations that

$$\int_{t \leq 4T/3} (S(t + \tfrac{h}{2}) - S(t - \tfrac{h}{2})) |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \ll \frac{1}{T} \sum_{n \leq L} |f(n)|^2,$$

and that

$$\int_{t > 5T/3} (S(t + \tfrac{h}{2}) - S(t - \tfrac{h}{2})) |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \ll \frac{1}{T} \sum_{n \leq L} |f(n)|^2.$$

Therefore, we obtain (3.3).

We extract extreme values of $\pm\{S(t + h) - S(t)\}$ by

$$\begin{aligned} & \pm \int_{4T/3 \leq t \leq 5T/3} (S(t + \tfrac{h}{2}) - S(t - \tfrac{h}{2})) |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \\ & \leq \sup_{T \leq t \leq 2T} \{\pm(S(t + h) - S(t))\} \int_{-\infty}^{\infty} |R(t)|^2 \Phi\left(\frac{t - 3T/2}{T/\log T}\right) dt \\ & \leq \frac{\sqrt{2\pi}T}{\log T} \left(1 + O\left(\frac{1}{T}\right)\right) \sup_{T \leq t \leq 2T} \{\pm(S(t + h) - S(t))\} \sum_{n \leq L} |f(n)|^2. \end{aligned}$$

In the last step, we have used Lemma 3.2. Combining this with (3.2), we obtain

$$\begin{aligned} & \sup_{T \leq t \leq 2T} \{ \pm(S(t+h) - S(t)) \} \times \sum_{n \leq L} |f(n)|^2 \\ & \geq \mp \left(1 + O\left(\frac{1}{T}\right) \right) \frac{2}{\pi} \operatorname{Re} \sum_{2 \leq k \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) \sum_{km \leq L} f(m) \overline{f(km)} + O\left(\frac{1}{T} \sum_{n \leq L} |f(n)|^2\right). \end{aligned}$$

This completes the proof of Proposition 2. $\square$

4. Proof of Theorem 1

In this section, we let L denote a large number, $h \in [C/\log L, c/\log \log L]$ with positive constants C large and c small. We choose $f = f_{\pm}$ as the multiplicative function supported on square-free numbers such that for any prime p

$$f_{\pm}(p) := \pm \sqrt{Q} \cdot \frac{\sin(\frac{h}{2} \log p)}{p^{1/2+\kappa h} h \log p}$$

if $\exp(\sqrt{\log \log L}/\sqrt{h}) =: M < p \leq L$ and $f_{\pm}(p) = 0$ otherwise. Here, the numbers κ and Q are to be chosen as

$$Q = 4\kappa(1-y)h \log L \Big/ \pi \int_0^{h \log L/2\pi} \frac{\sin^2(\pi u)}{(\pi u)^3} \frac{e^{2\kappa\pi u} - 1}{e^{4\kappa\pi u}} du, \quad \kappa = \frac{\log(h \log L)}{yh \log L},$$

where $y = \sqrt{\log(h \log L)/h \log L}$. It then holds that $Q \asymp h \log L$ and κ, y are sufficiently small when $h \in [C/\log L, c/\log \log L]$. These parameters are determined to optimize quantity (4.2) below. For this $f_{\pm}$, we give a lower bound of the ratio of resonator in the following proposition.

Proposition 4. *Let L be large, and let $C/\log L \leq h \leq c/\log \log L$ with positive constants C large and c small. Then we have*

$$\begin{aligned} & \pm \frac{2}{\pi} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) f_{\pm}(m) f_{\pm}(km) \Big/ \sum_{n=1}^{\infty} f_{\pm}(n)^2 \\ & \geq \left\{ 1 + O\left(\sqrt{h \log \log L} + \min \left\{ \sqrt{\frac{\log^3(h \log L)}{h \log L}}, \frac{(\log \log L)^{3/2}}{h^{3/2} \log L} \right\} \right) \right\} \sqrt{\frac{h}{\pi} \log L}. \end{aligned}$$

Theorem 1 immediately follows from Proposition 2 and this proposition in the case $L = T/(\log T)^2$.

Proof. Put $\alpha = \kappa h$. First, we observe by the definition of $f_{\pm}$ that

$$\pm \frac{2}{\pi} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) f_{\pm}(m) f_{\pm}(km) = \sqrt{Q} \frac{2}{\pi} \sum_{M < p \leq L} \frac{\sin^2(\frac{h}{2} \log p)}{p^{1+\alpha} h \log p} \sum_{\substack{m \leq L/p \\ p \nmid m}} f_{\pm}(m)^2. \quad (4.1)$$

We find by the definition of $f_{\pm}$ and Rankin's trick that

$$\begin{aligned}
\sum_{\substack{m \leq L/p \\ p \nmid m}} f_{\pm}(m)^2 &\geq \sum_{\substack{n=1 \\ p \nmid n}}^{\infty} f_{\pm}(n)^2 - \left(\frac{p}{L}\right)^{\alpha} \sum_{\substack{n=1 \\ p \nmid n}}^{\infty} f_{\pm}(n)^2 n^{\alpha} \\
&= \prod_{\substack{M < q \leq L \\ q \neq p}} (1 + f_{\pm}(q)^2) - \left(\frac{p}{L}\right)^{\alpha} \prod_{\substack{M < q \leq L \\ q \neq p}} (1 + f_{\pm}(q)^2 q^{\alpha}) \\
&= \frac{1}{1 + f_{\pm}(p)^2} \prod_{M < q \leq L} (1 + f_{\pm}(q)^2) - \left(\frac{p}{L}\right)^{\alpha} \frac{1}{1 + f_{\pm}(p)^2 p^{\alpha}} \prod_{M < q \leq L} (1 + f_{\pm}(q)^2 q^{\alpha}) \\
&= \left(\frac{1}{1 + f_{\pm}(p)^2} - \frac{(p/L)^{\alpha}}{1 + f_{\pm}(p)^2 p^{\alpha}} \prod_{M < q \leq L} \frac{1 + f_{\pm}(q)^2 q^{\alpha}}{1 + f_{\pm}(q)^2} \right) \times \prod_{M < q \leq L} (1 + f_{\pm}(q)^2).
\end{aligned}$$

Since the estimate $f_{\pm}(p)^2 p^{\alpha} \ll Q/\log L \asymp h$ holds and $f_{\pm}$ is supported on square-free and $M < p \leq L$, this is also equal to

$$\left\{ 1 - \left(1 + O\left(\frac{Q}{\log L}\right) \right) \left(\frac{p}{L}\right)^{\alpha} \prod_{M < q \leq L} \frac{1 + f_{\pm}(q)^2 q^{\alpha}}{1 + f_{\pm}(q)^2} + O\left(\frac{Q}{\log L}\right) \right\} \sum_{n=1}^{\infty} f_{\pm}(n)^2.$$

Observe that

$$\begin{aligned}
\prod_{M < q \leq L} \frac{1 + f_{\pm}(q)^2 q^{\alpha}}{1 + f_{\pm}(q)^2} &= \prod_{M < q \leq L} \left(1 + \frac{f_{\pm}(q)^2 (q^{\alpha} - 1)}{1 + f_{\pm}(q)^2} \right) \\
&= \exp \left(\left(1 + O\left(\frac{Q}{\log L}\right) \right) \sum_{M < q \leq L} f_{\pm}(q)^2 (q^{\alpha} - 1) \right).
\end{aligned}$$

Routine calculations using the prime number theorem and partial summation show that

$$\begin{aligned}
\sum_{M < p \leq L} \frac{\sin^2(\frac{h}{2} \log p)}{p^{1+\alpha} h \log p} &= \frac{\pi}{2} \varphi_2(\mathcal{L}; \kappa) + O\left(\sqrt{h \log \log L}\right), \\
\sum_{M < q \leq L} f_{\pm}(q)^2 (q^{\alpha} - 1) &= \left(1 + O\left(\sqrt{h \log \log L}\right) \right) \frac{\pi}{4} Q \varphi_3(\mathcal{L}; \kappa),
\end{aligned}$$

and that

$$\sum_{M < p \leq L} \frac{\sin^2(\frac{h}{2} \log p)}{p h \log p} = \frac{\pi}{2} \varphi(\mathcal{L}) + O\left(\sqrt{h \log \log L}\right),$$

where $\mathcal{L} = \frac{h}{2\pi} \log L$, and φ is as in the statement of Theorem 3, and φ_2, φ_3 are by

$$\begin{aligned}
\varphi_2(\mathcal{L}; \kappa) &:= \int_0^{\mathcal{L}} \left(\frac{\sin(\pi u)}{\pi u} \right)^2 e^{-2\pi \kappa u} du, \\
\varphi_3(\mathcal{L}; \kappa) &:= \int_0^{\mathcal{L}} \frac{\sin^2(\pi u)}{(\pi u)^3} \frac{e^{2\pi \kappa u} - 1}{e^{4\pi \kappa u}} du.
\end{aligned}$$

Therefore, quantity (4.1) is

$$\geq \left\{ \varphi_2(\mathcal{L}; \kappa) - (1 + E_1) \varphi(\mathcal{L}) \exp\left(-\kappa h \log L + (1 + E_2) \frac{\pi}{4} Q \varphi_3(\mathcal{L}; \kappa)\right) + E_3 \right\} \sqrt{Q}. \quad (4.2)$$

Here, the error terms E_1, E_2, E_3 satisfy $E_1 \ll h$, and $E_2, E_3 \ll \sqrt{h \log \log L}$. By the choice of Q , we find that this lower bound is

$$\{\varphi_2(\mathcal{L}; \kappa) - (1 + E_1)\varphi(\mathcal{L}) \exp(-\kappa(y - (1 - y)E_2)h \log L) + E_3\} \sqrt{\frac{4\kappa(1 - y)}{\pi\varphi_3(\mathcal{L}; \kappa)}} \sqrt{h \log L}. \quad (4.3)$$

Noting the choices of y and κ , we see that for $l = 1/\kappa \sqrt{\log(1/\kappa)}$

$$\begin{aligned} \varphi_2(\mathcal{L}; \kappa) &= \int_0^l \left(\frac{\sin(\pi u)}{\pi u} \right)^2 (1 + O(\kappa u)) du + O\left(\int_l^{\mathcal{L}} \frac{du}{u^2} \right) \\ &= \int_0^\infty \left(\frac{\sin(\pi u)}{\pi u} \right)^2 du + O\left(\kappa \log l + \frac{1}{l} \right) = \frac{1}{2} + O(\kappa \log(1/\kappa)), \end{aligned}$$

and that

$$\begin{aligned} \varphi_3(\mathcal{L}; \kappa) &= 2\kappa \int_0^l \left(\frac{\sin(\pi u)}{\pi u} \right)^2 (1 + O(\kappa u)) du + O\left(\int_l^{\mathcal{L}} \frac{du}{u^3} \right) \\ &= 2\kappa \int_0^\infty \left(\frac{\sin(\pi u)}{\pi u} \right)^2 du + O\left(\kappa^2 \log l + \frac{1}{l^2} \right) = \kappa(1 + O(\kappa \log(1/\kappa))). \end{aligned}$$

Hence, (4.3) is

$$\left\{ 1 + O\left(\sqrt{h \log \log T} + \min\left\{ \sqrt{\frac{\log^3(h \log T)}{h \log T}}, \frac{(\log \log T)^{3/2}}{h^{3/2} \log T} \right\} \right) \right\} \sqrt{\frac{h}{\pi} \log L},$$

which completes the proof of (4.1). $\square$

The above proof gives a good lower bound of the ratio of resonators. In particular, we obtain the following theorem by combining the lower bound with Theorem 3.

Theorem 4. *Let $\mathcal{A}$ be the set of arithmetic functions such that the value at one is not equal zero. For any large L and for $h \in [C/\log L, c/\log \log L]$ with positive constants C large and c small, we have*

$$\sup_{f \in \mathcal{A}} \left\{ \pm \operatorname{Re} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) f(m) \overline{f(km)} \middle/ \sum_{n \leq L} |f(n)|^2 \right\} = (1 + E) \sqrt{\frac{h}{\pi} \log L},$$

where

$$E \ll \min\left\{ \sqrt{\frac{\log^3(h \log L)}{h \log L}}, \frac{(\log \log L)^{3/2}}{h^{3/2} \log L} \right\} + \sqrt{h \log \log L}.$$

Proof. The lower bound has already shown in (4.1). The upper bound can be also proved by Theorem 3. Actually, we use Theorem 3 with $W = \frac{4}{\pi} h \log L$ to obtain that

$$\begin{aligned} &\left| \operatorname{Re} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) f(m) \overline{f(km)} \right| \middle/ \sum_{n \leq L} |f(n)|^2 \\ &\leq \frac{\varphi\left(\frac{h}{2\pi} \log L\right)}{2} \sqrt{\frac{4}{\pi} h \log L} + \frac{h \log L}{\pi \sqrt{\frac{4}{\pi} h \log L}} + O(h \sqrt{h \log L}) \leq (1 + O(h)) \sqrt{\frac{h}{\pi} \log L} \end{aligned}$$

since $\varphi(x) \leq 1/2$. Thus, we also obtain the upper bound. $\square$

5. Proof of Theorem 2

Let r be a sufficiently large positive integer. For any $\theta' > 0$ and any large $T \geq T_0(r, \theta')$, we have

$$\begin{aligned} \inf_{T \leq t \leq 2T} S(t + 2\pi r \theta' / \log T) - S(t) &\leq - \left(1 + O \left(\frac{(\log r)^{3/2}}{r^{1/2}} \right) \right) \sqrt{2r\theta'} \\ &= -r \left(\sqrt{\frac{2}{r}} \theta' + O \left(\frac{(\log r)^{3/2}}{r} \right) \right) \end{aligned}$$

by Theorem 1. Therefore, if θ is chosen as

$$\theta = 1 + \frac{\sqrt{2}}{\sqrt{r}} - C_1 \frac{(\log r)^{3/2}}{r}$$

with C_1 a sufficiently large absolute positive constant, then the inequality

$$\inf_{T \leq t \leq 2T} \{S(t + 2\pi r \theta' / \log T) - S(t)\} \leq -b$$

holds for any large T , where

$$\theta' = 1 + \frac{\sqrt{2}}{\sqrt{r}} - \frac{C_1}{2} \frac{(\log r)^{3/2}}{r} > \theta,$$

and

$$b = r(\theta' - 1) + \frac{C_1}{3} (\log r)^{3/2} > r(\theta' - 1).$$

Hence, by Proposition 1, we obtain the inequality of λ_r in (1.4). Similarly, we can prove the inequality of μ_r . $\square$

6. Proof of Theorem 3

Let h be an arbitrary positive number, and let L be large. Let W be an arbitrary positive number. Define $g(n) = \sqrt{W}/h\sqrt{n} \log n$. Then we adapt Soundararajan's method, which uses an inequality similar to $2|f(m) \sin(\frac{h}{2} \log k) f(km)| \leq |f(m)|^2 \sin^2(\frac{h}{2} \log k) g(k) + |f(km)|^2/g(k)$, to obtain

$$\begin{aligned} &\left| \operatorname{Re} \frac{2}{\pi} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) f(m) \overline{f(km)} \right| \\ &\leq \frac{1}{\pi} \sum_{2 \leq k \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sum_{km \leq L} \left(|f(m)|^2 \sin^2\left(\frac{h}{2} \log k\right) g(k) + \frac{|f(km)|^2}{g(k)} \right) \\ &= \frac{1}{\pi} \sum_{n \leq L} |f(n)|^2 \left(\sqrt{W} \sum_{p^a \leq L/n} \frac{\sin^2(\frac{h}{2} \log p^a)}{l^2 p^a h \log p} + \sum_{k|n} \frac{h \Lambda(k)}{\sqrt{W}} \right) \\ &= \frac{1}{\pi} \sum_{n \leq L} |f(n)|^2 \left(\sqrt{W} \sum_{p^a \leq L/n} \frac{\sin^2(\frac{h}{2} \log p^a)}{a^2 p^a h \log p} + \frac{h \log n}{\sqrt{W}} \right). \end{aligned}$$

Routine calculations using the prime number theorem and partial summation show that

$$\begin{aligned} \sum_{p^a \leq L/n} \frac{\sin^2(\frac{h}{2} \log p^a)}{a^2 p^a h \log p} &= \sum_{p \leq L/n} \frac{\sin^2(\frac{h}{2} \log p)}{p h \log p} + O(h) \\ &= \int_2^{L/n} \frac{\sin^2(\frac{h}{2} \log \xi)}{\xi h (\log \xi)^2} d\xi + O(h) = \frac{\pi}{2} \varphi\left(\frac{h}{2\pi} \log(L/n)\right) + O(h). \end{aligned}$$

Therefore, it holds that

$$\begin{aligned} &\frac{1}{\pi} \sum_{n \leq L} |f(n)|^2 \left\{ \sqrt{W} \sum_{p^a \leq L/n} \frac{\sin^2(\frac{h}{2} \log p^a)}{l^2 p^a h \log p} + \frac{h \log n}{\sqrt{W}} \right\} \\ &\leq \frac{1}{\pi} \left(\max_{n \leq L} \left\{ \sqrt{W} \frac{\pi}{2} \varphi\left(\frac{h}{2\pi} \log(L/n)\right) + \frac{h \log n}{\sqrt{W}} \right\} + O(h\sqrt{W}) \right) \sum_{n \leq L} |f(n)|^2 \\ &\leq \left(\max_{1 \leq l \leq L} \left\{ \frac{\sqrt{W}}{2} \varphi\left(\frac{h}{2\pi} \log(L/l)\right) + \frac{h \log l}{\pi \sqrt{W}} \right\} + O(h\sqrt{W}) \right) \sum_{n \leq L} |f(n)|^2. \end{aligned}$$

Hence, we have

$$\begin{aligned} &\left| \operatorname{Re} \frac{2}{\pi} \sum_{km \leq L} \frac{\Lambda(k)}{\sqrt{k} \log k} \sin\left(\frac{h}{2} \log k\right) f(m) \overline{f(km)} \right| \\ &\leq \left(\max_{1 \leq l \leq L} \left\{ \frac{\sqrt{W}}{2} \varphi\left(\frac{h}{2\pi} \log(L/l)\right) + \frac{h \log l}{\pi \sqrt{W}} \right\} + O(h\sqrt{W}) \right) \sum_{n \leq L} |f(n)|^2. \end{aligned}$$

Since the parameter W is arbitrary, we have (1.5). $\square$

7. Limitations of λ_1 and μ_1 deduced from Theorem 3

If $\tau(\xi; f) > 1$ for any $L \leq T$, any arithmetic function f , and any $\xi \geq \xi_0$, then the bound $\lambda_1 \geq \xi_0$ becomes the limitation for the Montgomery-Odlyzko method. Note that the right hand side of the inequality in Theorem 3 is increasing for L , and hence we may assume $L = T$ in the following argument. Using Theorem 3 with $l = T^x$, $W = 22.6$, we have

$$\tau(\xi; f) \geq \xi - \max_{0 \leq x \leq 1} \left\{ \frac{\sqrt{W}}{2} \varphi(\xi(1-x)) + \frac{2\xi x}{\sqrt{W}} \right\} - o(1)$$

for any arithmetic function f that is not identically zero. The right hand side exceeds one when $\xi \geq \xi_0 = 3.022$. From this observation, we deduce that the limitation of λ_1 for the Montgomery-Odlyzko method is $\lambda_1 \geq 3.022$. Similarly, we conclude that the limitation of μ_1 for the Montgomery-Odlyzko method is $\mu_1 \leq 0.508$ by using Theorem 3 with $W = 4.9$.

Remark 2. We should mention the difference between our work and Goldston, Trudgian, Turnage-Butterbaugh [6]. The works [4], [6] used the inequality $|zw| \leq (|z|^2 + |w|^2)/2$ for $w, z \in \mathbb{C}$, partial summation, and the inequality $|\sin x| \leq |x|$ for $x \in \mathbb{R}$. On the other hand, we avoid using the final inequality and instead derive the result by a direct analysis of the sine function.

Acknowledgments. The authors would like to thank Professor Timothy Trudgian for providing us with valuable comments. The first author is supported by JSPS KAKENHI Grant Number 24K16907. The second author is supported by JSPS KAKENHI Grant Number

25K17245. The third author is supported by Grant-in-Aid for JSPS Research fellow Grant Number 24KJ1235.

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